

Week 3+ : Warmup exercises

! Week 3 - 6/7 : Snt CELE MAI DE BAZĂ !

↳ ec. dreptii
↳ ec. plomului

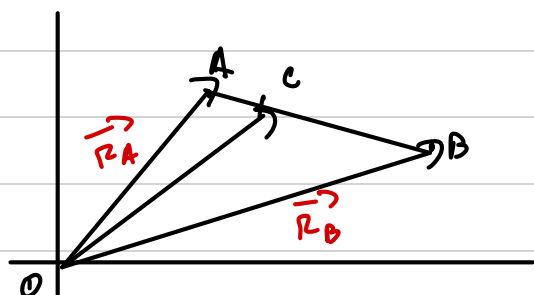
+ mai pe la final chestii cu curbe, chestii care se leagă de analiză

• mai e limită superioară de puncte neapăsător (1 pt max la mla finală)

• recuperare seminar 0 - online, probă sept. oct.

1.1. Let $\lambda = \frac{CA}{CB}$ - ratio in which $C \in [AB]$ divides it.

$$\text{Proof: } \vec{OC} = \frac{1}{1+\lambda} \vec{OA} + \frac{\lambda}{1+\lambda} \vec{OB}$$



$$\vec{OC} = \vec{OA} + \vec{AC}$$

$$\lambda = \frac{AC}{CB} \Rightarrow AC = \lambda CB \Rightarrow \vec{AC} = \lambda \cdot \vec{CB} \quad | + \vec{OA}$$

$$\Rightarrow (\lambda + 1) \vec{AC} = (\vec{CB} + \vec{AC}) \\ = (\lambda + 1) \cdot \vec{AB}$$

$$\begin{aligned}
 \Rightarrow \vec{OC} &= \vec{OA} + \lambda \vec{AB} = \\
 &= \frac{\lambda}{\lambda+1} (\vec{AO} + \vec{OB}) + \vec{OA} \\
 &= \frac{\lambda}{\lambda+1} \vec{OB} + \left(1 - \frac{\lambda}{\lambda+1}\right) \vec{OA} \\
 &= \frac{\lambda}{\lambda+1} \vec{OB} + \frac{1}{\lambda+1} \vec{OA}
 \end{aligned}$$

Alternatively : (with pos. vectors)

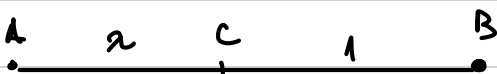
$$\vec{AC} = \lambda \vec{CB}$$

$$\vec{OC} - \vec{OA} = \lambda (\vec{OB} - \vec{OC})$$

$$(\lambda+1)\vec{OC} = \vec{OA} + \lambda \vec{OB}$$

$$\vec{OC} = \frac{\vec{OA} + \lambda \vec{OB}}{1 + \lambda}$$

Nice's way - Trick 4 memory



← not on the Exam paper

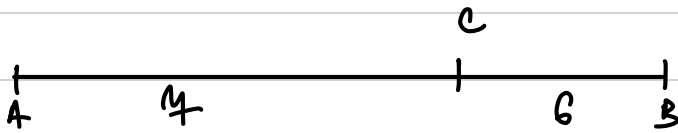
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$$\begin{aligned}
 \lambda \vec{CB} + \vec{AC} &= (\lambda+1)\vec{OC} \\
 \vec{OC} &= \frac{\vec{AC} + \lambda \vec{CB}}{\lambda+1}
 \end{aligned}$$

Ex: $C \in [AB]$

$$\frac{AB}{BC} = \frac{13}{6}$$



$$\vec{r}_C = \frac{4\vec{r}_B + 6\vec{r}_A}{13}$$

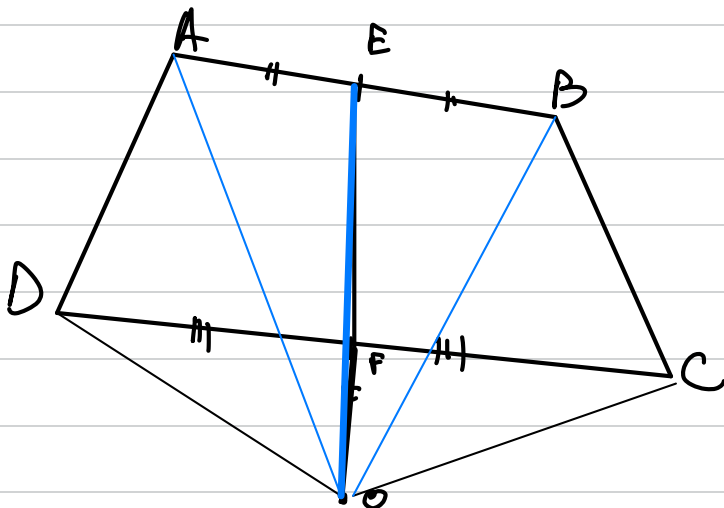
Reversing the roles

1.3. ABCD quadrilateral

E mid point of [AB]

F mid point of [CD]

$$\vec{EF} = \frac{1}{2} (\vec{AD} + \vec{BC})$$



$$\vec{EF} = \vec{r}_D - \vec{r}_E$$

$$\vec{r}_D = \frac{\vec{r}_D + \vec{r}_C}{2}$$

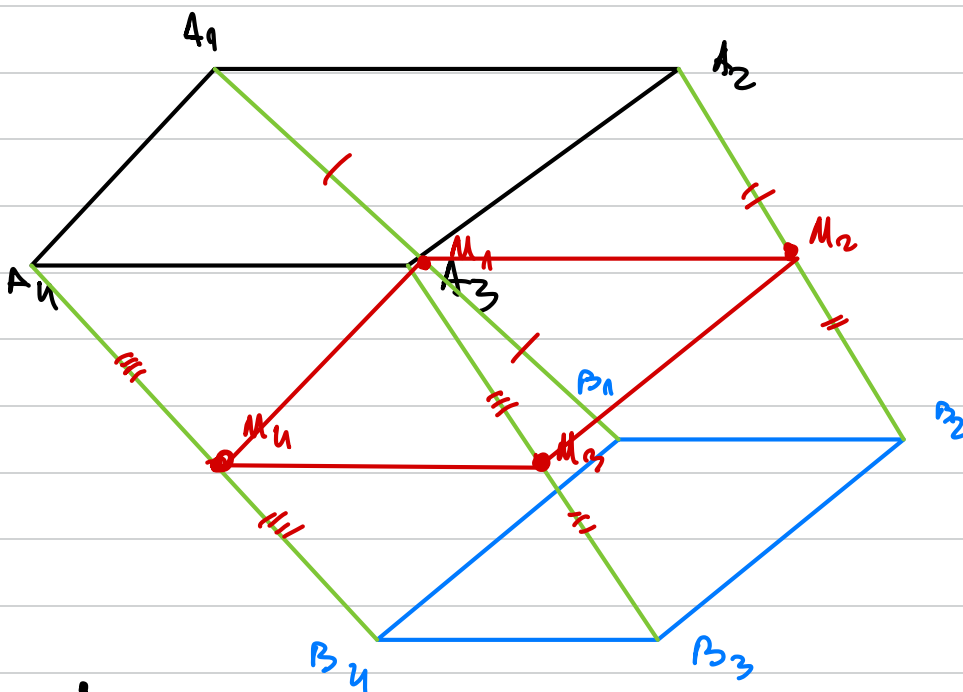
$$\vec{r}_E = \frac{\vec{r}_B + \vec{r}_A}{2}$$

$$\begin{aligned}
 \vec{EP} &= \frac{\vec{r_D} + \vec{r_C}}{2} - \frac{\vec{r_B} + \vec{r_A}}{2} \\
 &= \frac{\vec{r_D} - \vec{r_A}}{2} + \frac{\vec{r_C} - \vec{r_B}}{2} \\
 &= \frac{1}{2} (\vec{AD} + \vec{BC})
 \end{aligned}$$

1.5. $\left. \begin{array}{l} A_1 A_2 A_3 A_4 \\ B_1 B_2 B_3 B_4 \end{array} \right\} \text{parallelog.}$

Show that the midpoints M_1, M_2, M_3, M_4 of $\{A_1 B_1\}$
 $\{A_2 B_2\}$
 $\{A_3 B_3\}$
 $\{A_4 B_4\}$

are vertices of a parallelogram



$$\vec{r_{M1}} = \frac{\vec{r_{A1}} + \vec{r_{B1}}}{2}$$

$$\vec{r_{M2}} = \frac{\vec{r_{A2}} + \vec{r_{B2}}}{2}$$

$$\vec{r_{M3}} = \frac{\vec{r_{A3}} + \vec{r_{B3}}}{2}$$

$$\vec{r_{M4}} = \frac{\vec{r_{A4}} + \vec{r_{B4}}}{2}$$

$$\overrightarrow{M_1 M_3} = \overrightarrow{M_1 M_2} + \overrightarrow{M_2 M_3} = \frac{1}{2} (\overrightarrow{A_1 A_3} + \overrightarrow{B_1 B_3})$$

$$A_1 A_2 A_3 A_4 - \text{parallelogram} \Rightarrow \overrightarrow{A_1 A_3} = \overrightarrow{A_1 A_2}$$

$$B_1 B_2 B_3 B_4 - \text{parallelogram} \Rightarrow \overrightarrow{B_1 B_3} = \overrightarrow{B_1 B_2}$$

$$\overrightarrow{M_1 M_3} = \frac{1}{2} (\overrightarrow{A_1 A_2} + \overrightarrow{B_1 B_2}) = \overrightarrow{M_1 M_2}$$

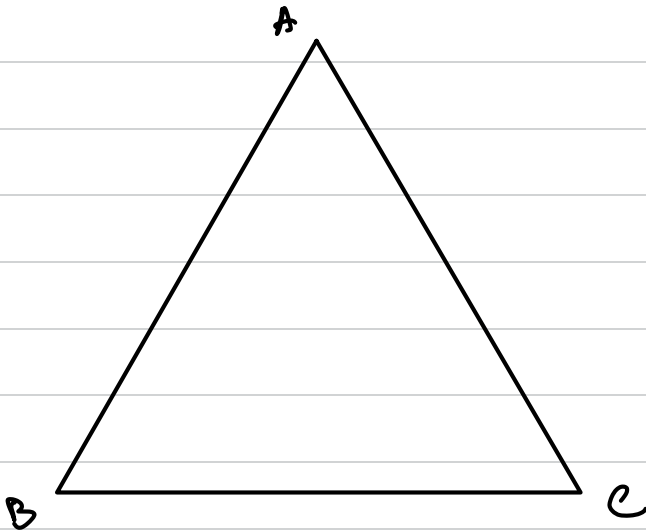
$\hookrightarrow$ same length
 $\hookrightarrow$ same direction $\Rightarrow$

$$\Rightarrow \overrightarrow{M_1 M_3} \parallel \overrightarrow{M_1 M_2} \Rightarrow M_1 M_2 M_3 M_4 - \text{parallelogram}$$

$\hookrightarrow \parallel \text{ AND } =$

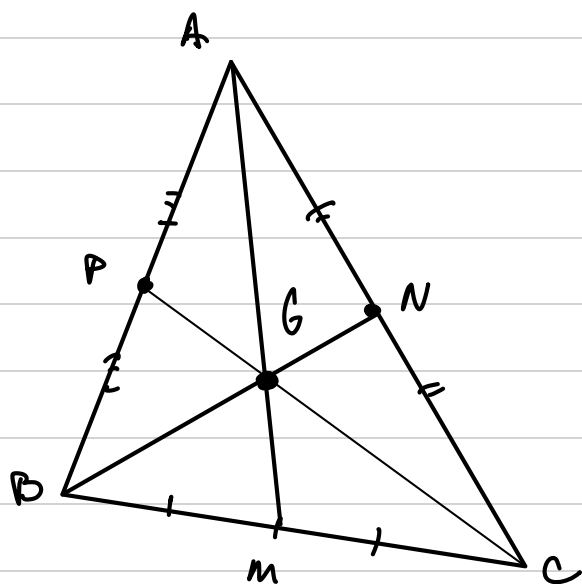
INTERLUDE

1.6. Starting from Hilbert's axioms prove that the base angles of an isosceles triangle are congruent.



$$\left. \begin{array}{l} AB = AC \\ AC = AB \\ \widehat{BAC} = \widehat{CAB} \end{array} \right\} \begin{array}{l} \text{By III.6} \\ \Rightarrow \widehat{ABC} = \widehat{ACB} \end{array}$$

1.4. Using vectors show that the medians in a Δ are concurrent and deduce the ratio in which the common intersection point divides the medians.



The vector eq. of a line through points A and B

$\forall t \in \mathbb{R} \exists ! \lambda \in \mathbb{R} :$

$$\vec{r}_T = \lambda \vec{r}_A + (1-\lambda) \vec{r}_B$$

Let M, N, P - midpoints of the sides

Let $\{G\} = AM \cap BN$

We will find $\vec{r}_G$ and prove $G \in CP$

$$\vec{BA} + \vec{BC} = 2 \vec{BN}$$

$$\vec{AC} + \vec{AB} = 2 \vec{AM}$$

$$\vec{BA} + \vec{BC} + \vec{AC} + \vec{AB} = 2(\vec{BN} + \vec{AM})$$

$$\vec{BC} + \vec{AC} = 2(\vec{BN} + \vec{AM}) \quad | \cdot (-1)$$

$$\vec{CB} + \vec{CA} = 2(\vec{NB} + \vec{MA})$$

$$2\vec{CP} = 2(\vec{NB} + \vec{MA})$$

$$\overrightarrow{CP} = \overrightarrow{NP} + \overrightarrow{MA}$$

$$\overrightarrow{CP} = \overrightarrow{NG} + \overrightarrow{GP} + \overrightarrow{MG} + \overrightarrow{GA}$$

Blockaj la totala + mental

$$\begin{aligned} \text{GG AM} \Rightarrow \vec{r}_G &= z \vec{r}_A + (1-z) \vec{r}_M \\ &= z \vec{r}_A + \frac{1-z}{2} \vec{r}_B + \frac{1+z}{2} \vec{r}_C \end{aligned}$$

$$\begin{aligned} \text{GG BN} \Rightarrow \vec{r}_G &= \mu \vec{r}_B + (1-\mu) \vec{r}_N \\ &= \mu \vec{r}_B + \frac{1-\mu}{2} \vec{r}_A + \frac{1+\mu}{2} \vec{r}_C \end{aligned}$$

$$\begin{aligned} \left(z - \frac{1-\mu}{2} \right) \vec{r}_A + \left(\frac{1+z}{2} - \mu \right) \vec{r}_B + \\ + \left(\frac{1-z}{2} - \frac{1+\mu}{2} \right) \vec{r}_C = 0 \end{aligned}$$

Fix the origin in A:

$$\vec{r}_A = \vec{0}, \vec{r}_B = \vec{AB}, \vec{r}_C = \vec{AC}$$

lin. independent

So :

$$\left(\frac{1-\lambda}{2} - \mu\right) \vec{AB} + \left(\frac{1-\lambda}{2} - \frac{1-\mu}{2}\right) \vec{AC} = 0$$

ΔABC is non-degenerate $\Rightarrow \vec{AB}, \vec{AC}$ lin. indep. $\Rightarrow$

Recap Base Changes from Lin. Alg.

$$\Rightarrow \begin{cases} \frac{1-\lambda}{2} - \mu = 0 \\ \frac{1-\lambda}{2} - \frac{1-\mu}{2} = 0 \end{cases} \Rightarrow \mu = \lambda$$

$$\Rightarrow 1-\lambda = 2\lambda \Rightarrow \lambda = \frac{1}{3}$$

$$\begin{aligned} \text{So: } \vec{r_G} &= \frac{1}{3} \vec{r_A} + \frac{1}{3} \vec{r_B} + \frac{1}{3} \vec{r_C} = \\ &= \frac{\vec{r_A} + \vec{r_B} + \vec{r_C}}{3} \end{aligned}$$

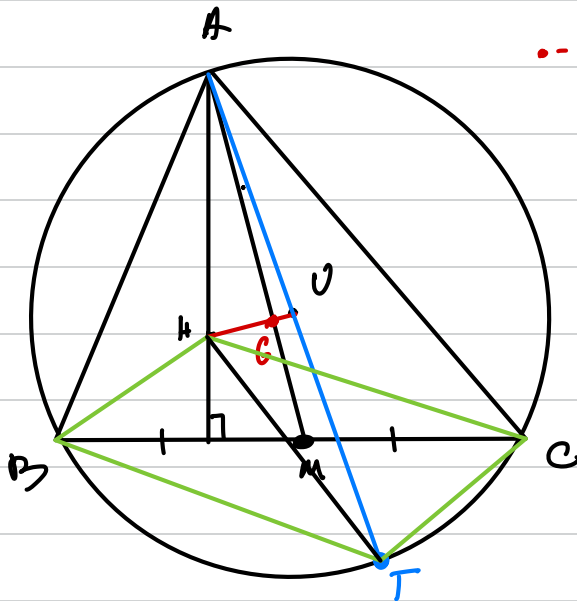
$$\vec{CG} = \vec{r_G} - \vec{r_C} = \frac{\vec{r_A} + \vec{r_B} - 2\vec{r_C}}{3}$$

$$\vec{CP} = \vec{r_P} - \vec{r_C} = \frac{\vec{r_A} + \vec{r_B}}{2} - \vec{r_C}$$

$$\overrightarrow{CP} = \frac{3}{2} \overrightarrow{CG} \Rightarrow C, P, G \text{ collinear} \Rightarrow$$

$$\Rightarrow G = AM \cap BN \cap CP \text{ and } CP = \frac{3}{2} CG$$

1.8. G, H, U collinear



-- Euler's line

? $HCTB$ - parallel?